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Exploring the factors influencing consumer engagement behavior regarding short-form video advertising: A big data perspective

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Highlights

- Performance expectancy, entertainment, and tie strength have positive effects on consumer engagement behavior.
- Sales approach has negative effect on consumer engagement behavior.
- The effect of performance expectancy on likes is stronger for search goods than for experience goods.
- The influence of entertainment and sales approach on comments is stronger for experience goods than for search goods.
- The influence of tie strength on comments is stronger for search goods than for experience goods.

Abstract

Short-form video has attracted users' attention and been widely adopted for entertainment. Recently, short-form video has also been used for advertising. However, how short-form video for advertisement influences consumer engagement behavior remains unclear. This study aims to explore key features of short-form video advertisements that influence consumer engagement behavior. Through analyzing data obtained from social media platform TikTok, we discovered that four key features of short-form video—performance expectancy, entertainment, tie strength, and sales approach—are significantly related to consumer engagement behavior. In addition, the results showed that product type moderated the relationship of these effects on consumer engagement behavior. This study is one of the first to investigate the influence of short-form video advertisement features on consumer engagement behavior; thus, it contributes to the social media advertisement literature. It extends consumer engagement behavior research by applying a combination of uses and gratifications theory and signal theory. It also highlights the significance of product type in advertising literature. The use of big data and text analysis contributes from a methodological perspective to social media research. This study also provides practical and managerial implications for sellers and marketers on how to attract consumers to engage in videos and how to make data-driven decisions.

Keywords

Short-form video advertisements; Consumer engagement behavior; Product-type; Text analysis; TikTok

Previous article in this issue Next article in this issue

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al media has provided a more convenient way for people to communicate with others and created a new
social commerce (s-commerce) (Busalim and Hussin, 2016). S-commerce can be implemented in two ways: (1) social network features added to traditional e-commerce or (2) commercial features added to social network sites (Liang and Turb

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has emerged only recently. Examples include the livestreaming commerce and short-form video commerce used in TikTok (or Douyin in Chinese) and KuaiShou. According to recent research (Iresearch, 2020b), the penetration rate of internet users and online shopping users is close to saturation and traditional e-commerce has entered a period of traffic mining. Short-form video commerce, a typical type of video content commerce, has become an emerging strategy for e-commerce firms to obtain and monetize users' traffic.

By December 2021, the number of short-video application users reached 934 million, accounting for 90.5% of the total netizens (CNNCI, 2022). Such vast traffic led many firms to invest in short-form video commerce. According to the report QuestMobile (2021), short-form video advertising revenue reached CNY 50.37 billion in the first half of 2021 (approximately USD 7.79 billion), ranking first among all types of advertising. In contrast to other marketing methods, short-form video advertising has two significant advantages. First, short-form video has natural benefits in increasing user stickiness and user traffic aggregation. Second, short-form video advertising costs less than traditional advertising in terms of brand building and information dissemination for firms (Iresearch, 2020b).

Despite its growing popularity and advantages, studies pertaining to short-form video advertising are still at a nascent stage. Previous studies have focused mainly on exploring the factors that influence users' participation in and addiction to short-form video applications. For example, Rubenking (2019) reported that strong emotional and arousal response experiences will increase users' intent to share disgusting online videos. Hu et al. (2016) revealed that self-construal and community interaction have positive effects on user participation in video-sharing communities. It has been found that gratification, narcissism, and personal traits have significant influences on customer engagement that includes contributions, enhancement, and creations (Fei et al., 2021). Owing to the excessive use of short-form video applications, Zhang et al. (2019) argued that interpersonal attachments, such as interaction anxiety and social isolation, and site attachments, including entertainment and personalization, have positive influences on short-form video application addiction. Tian et al. (2022) revealed that users' perceived enjoyment and feelings of withdrawal are positively associated with addiction. There are some limitations to these studies. First, the commercial value of short videos has been largely overlooked. In addition, few studies have explored which features of short-form video content can influence users' engagement behavior (Fei et al., 2021; Hu et al., 2016; Yu et al., 2022). Without an understanding of these issues, it is impossible for video producers to produce attractive video content that effectively promotes relevant products.

Moreover, most studies in the short-form video literature have collected data using survey methods (Fei et al., 2021; Yang et al., 2021; Zhang et al., 2019), making it difficult to capture a wider set of opinions from consumers and real consumer behavior manifestations owing to the limitations of the survey, such as social desirability. The increasingly available big data provides opportunities to address this problem. Big data encompasses both non-textual data (e.g., voice, images, and videos) and textual data (e.g., comments and posts) captured through social media created by sellers and consumers (Erevelles et al., 2016; Liu et al., 2021). The application of unstructured data can extend prior studies through analysis of the various types of activities of sellers and consumers, and address the issue of survey data (Liu et al., 2021).

To address the gaps in the literature, this study examined how the features of short-form video advertisements can influence consumers' engagement behavior. We collected raw data of video and user interaction records from the TikTok platform and applied textual analysis to analyze the data, which allowed us to obtain a more persuasive and systematic result.

The remainder of this paper is structured as follows. First, the literature on short-form video commerce is reviewed, followed by an introduction to uses and gratifications (U&G) theory and signal theory, which is the theoretical basis of this study. The research model and hypotheses are then presented. Next, the data collection procedures and data analysis procedures are outlined, accompanied by the results of the data analysis. The key findings, along with the theoretical and practical implications, are further discussed. Finally, we highlight the limitations of current research and discuss future research directions with the conclusion for this study.

2. Literature review

2.1. Short-form video commerce

Short-form video, which lasts from a few seconds to a few minutes, is one of the most popular video formats (Iresearch, 2017; Wang, 2020). Users can create and share their videos, and interact with others through short-form video applications. For instance, users can share interesting experiences or useful tips on daily life. Users can click "like," comment, and share videos posted by other users, and follow the creators they like. The rich social interactions and vast user traffic provide opportunities for sellers to promote their products in short-form videos. This marketing model is called short-form video commerce, and videos with product promotion are considered short-form video advertising.

TikTok is a popular short-form video application. TikTok introduced the shopping cart (or the "little yellow car") that is embedded in the product window" functions for users in 2018 (Iresearch, 2020b). These functions have provided convenience for the powerful editing and customization functions of short-form video, sellers are able to promote their goods in can interact with sellers by commenting, liking, and sharing if the video meets their personal needs. Consumers ned to the video can click the "little yellow car" shortcut to jump to a third-party e-commerce platform or

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pulling grass. Fig. 1 shows the details of short-form video advertising.

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Fig. 1. The short-form video advertising interface: a) main video interface; b) comment interface.

With the popularity of short-form video among internet users, researchers have begun to investigate users' behavior in the short-form video context (see Table 1). As with other new technology, most studies have tried to investigate the factors that influence users' participation/adoption/engagement behavior in short-form video. For instance, Hu et al. (2016) discovered that effective self-construal (independent and interdependent construal) and community interactivity (active control and reciprocal communication) can promote user participation in video-sharing communities. Wang (2020) reported that the first-person camera view elevated viewers' immersion, whereas social presence, entertainment, and humor had mixed effects on interactions between users and mobile applications. Scherr and Wang (2021) revealed that socially rewarding self-presentation, trendiness, escapist addiction, and novelty are four distinct gratifications for TikTok use. Zhang et al. (2022) observed that appropriate combinations of common themes, music genres, and sentiment can strengthen the involvement and awareness of short-form video entertainment.

Table 1. The differences between our work and the short-form video literature.

Authors	Antecedents	Outcomes	Social or commercial roles	Users' motivations and opinions	Theory/model	Research approach
Hu et al. (2016)	• Self-construal (independent construal and interdependent construal) • Community interactivity (active control and reciprocal communication)	User participation	Social	✓	Triadic reciprocal determinism	Online survey
Zhang et al. (2019)	• Social system (social [illegible]ocial isolation) [illegible]tem (personalization,	Short-form video addiction	Social	✓	Attachment theory; Socio-technical approach	Online survey

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			commercial roles	motivations and opinions		approach
	entertainment) • Interpersonal attachment • Site attachment					
Rubenking (2019)	• Emotional response • Great arousal	Sharing intention	Social	✓	Theory of reasoned action	Online survey
Wang (2020)	• Use of humor • Camera view • Immersion • Spatial presence • Social presence • Perceptual realism • entertainment	Intent to adopt	Social		Human-Computer Interaction theories	Experimental design
Fei et al. (2021)	• Gratifications-sought • Narcissism • traits	Engagement behavior	Social	✓	U&G theory	Online survey

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			commercial roles	motivations and opinions		approach
Scherr and Wang (2021)	• Socially rewarding self-presentation • Trendiness • Escapist addiction • Novelty	TikTok use	Social	✓	Gratification niche theory	Online survey
Mulier et al. (2021)	• Mobile video advertisement format • Effort of watching the video advertisement on the smartphone	Consumer engagement	Commercial			Experimental design
Li et al. (2021)	• Personalization recommendation systems • Perceived information narrowing • Privacy concern • Psychological reactance	Discontinuous usage behavior	Social		Psychological Reactance Theory	Online survey
Liu et al., 2021, Liu et al., 2021, Liu et al., 2021	• Perceived stress • Self-compensation motivation	Short-form video addiction	Social			Survey
	nt • Convenience	Participation in viral social media challenges on TikTok	Social	✓	U&G theory	Semi-structured interviews

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			commercial roles	motivations and opinions		approach
	• Escape • Information • Social support • Socialization • Other factors beyond U&G theory					
Tian et al. (2022)	• Immersion • Social features • Control features • Feeling of withdrawal • Perceived enjoyment	Short-form video addiction	Social		The opponent process theory; Stimulus–organism–response framework	Online survey
Yiyi et al. (2022)	• Short video use • Body image	Weight loss intentions	Social	✓		Survey
Yu et al. (2022)	• Perceived ease of use • Perceived information • nce	• Usage time • Usage frequency • Degree of involvement	Social	✓	Motivation theories	Survey and in-depth interview

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			commercial roles	**motivations and opinions**		**approach**
	• Social interaction					
Zhang et al. (2022)	• Video characteristics	Citizen engagement	Social			Text analysis
Yuan et al. (2022)	• In-feed advertising	Online traffic	Commercial			Empirical Research
	• Key opinion leader endorsement					
This research	• **Signal of utilitarian value (performance expectancy)**	**Consumer engagement behavior manifestation**	**Social and commercial**	✓	**U&G theory; Signal theory**	**Big data and Text analysis**
	• **Signal of hedonic value (entertainment)**					
	• **Signal of social value (tie strength, sales approach)**					

Other researchers have noticed users' addictive behavior in response to short-form video apps and explored the various factors that may induce addiction behavior. For instance, Zhang et al. (2019) found that interpersonal attachment (social isolation and interaction anxiety) and site attachment (personalization and entertainment) had significant positive influences on short-form video app addiction. Liu et al. (2021) reported that perceived stress had a positive effect on short-form video addiction and this relationship was moderated by shyness. Tian et al. (2022) found that short-form video features (i.e., immersion features, social features, control features) influenced users' addiction by activating their perceived enjoyment and feelings of withdrawal, and that user's procrastination moderated the relationship between feelings of withdrawal and addiction.

Although these studies have provided some insights into user behaviors in the short-form video context, there are still two gaps in the literature. First, researchers have only paid attention to the social role of short-form video in influencing user behavior (Fei et al., 2021; Hu et al., 2016; Zhang et al., 2019, 2022); the commercial role of short-form video has been largely ignored. Although Mulier et al. (2021) revealed that mobile vertical video advertisements increase consumer interest and engagement when compared with mobile horizontal video advertisements, they did not provide a clear definition of mobile video advertising. The most relevant study may come from Yuan et al. (2022), who explored the effect of in-feed advertising and key opinion leader strategies in attracting product traffic for online retailors at TikTok. However, specific short-video advertising features have not been studied.

Second, most of these studies apply online surveys (Busalim and Hussin, 2016; Fei et al., 2021) or have an experimental design (Mulier et al., 2021; Wang, 2020) as their research method to examine user behavior; such designs are subject to several shortcomings, such as small sample sizes, the recall bias of respondents, and the inability to control the investigation environment. Users generate a large amount of behavioral data on short- reflect consumers' real reactions to short-form video advertisements. By analyzing this data, we could avoid the and experimental design.

aims to explore the factors influencing consumer engagement behavior regarding short-form video advertising ory.

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U&G theory originated in the 1940s as a reaction to traditional mass communication research, which emphasized the use of media to gratify users' various needs and wants—as emanating from the individual's social environment and serving as the motivation for using the media (Katz et al., 1974). It also referred to as "needs and gratifications theory," and is regarded as one of the most influential theories in media research. U&G theory focuses on explaining the social and psychological motives shaping peoples' use of media (Katz et al., 1974; Rubin, 1994). It has three major objectives: to explain how media is used by individuals to satisfy their needs, to understand the motives for media use, and to identify the outcomes that follow from needs, motives, and media use (Katz et al., 1973).

There are some essential assumptions in U&G theory: users are active and goal-directed and take the initiative to select and use communication tools to satisfy their needs, users are aware of their personal needs and find the appropriate media form to gratify them, and users' selection of a type of media is mainly driven by its ability to meet their needs (Katz et al., 1973; Palmgreen et al., 1981). These assumptions reveal that U&G theory explores users' various motivations for using media and thereby can provide a comprehensive research framework for studying customer motivations and behaviors from consumers' perspectives. U&G theory fits especially well with research on users' motivations and the voluntary use of media (Xiao, 2018). From this perspective, it is appropriate for us to apply U&G theory to investigate the factors influencing customer engagement behavior regarding short-form video advertising.

Prior studies have applied U&G theory to study the motivations of consumers that influence their behavior. For example, on the basis of U&G theory, Verhagen et al. (2015) observed that cognitive benefits, social integrative benefits, personal integrative benefits, and hedonic values positively influence customer engagement intentions. Busalim et al. (2021) applied U&G theory to explore the impact of social interaction, technological factors, and motivational factors on customer engagement behavior in social commerce platforms. On the basis of U&G theory, Kujur and Singh (2017) discovered that the content characteristics of social networking advertisements, such as entertainment, information, and interactivity, have direct influences on consumers' participation.

In general, there are three main types of motivations that influence consumers' behavior: utilitarian values ("How much information can I get when I engage in the social media and how will the product meet my needs?"), hedonic values ("How will the social media bring me entertainment and relaxation?"), and social values ("How can the content provide me with a convenient platform for meeting and making more friends?") (Wongkitrungrueng and Assarut, 2020). These are also the main motivations for consumers to engage in short-form video advertisements, which provide a basic frame through which to explore the factors that influence consumer engagement behavior in the short-form video advertising context.

2.3. Signaling theory

According to Spence (1978), information asymmetry and misinterpretation may exist between the sender and receiver in the process of information transmission. Generally, those who send information clearly understand their personal situations, whereas receivers may not (Mavlanova et al., 2012). In this situation, signaling theory can provide an effective framework for senders to convey useful signals to receivers (Wang et al., 2019). Signals could reflect the objective situation of senders and thereby have a significant influence on receivers' cognitive perceptions and behaviors (Wells et al., 2011).

A signal, to be effective, should meet two conditions: (1) from the view of senders, signals need to be sufficiently costly to distinguish the sender from others; (2) from the view of receivers, signals have to clearly reflect the real quality of the source, emphasizing the importance of receivers' sights. Previous studies have used signaling theory in research on paid knowledge products (Cai et al., 2020), e-commerce (Mavlanova et al., 2012; Meents and Verhagen, 2018), and health-related trust (Hampshire et al., 2017). However, most of these studies are conducted from the perspectives of senders, which ignores the cognition of receivers and may not lead to a comprehensive conclusion. To solve this problem, we applied signaling theory in combination with U&G theory. Signaling theory offers clear conditions for effective signals, thereby providing an appropriate framework for short-form video producers to understand what factors consumers really care about. U&G theory fully considers receivers' perceptions, thereby providing consumers with real motivations for their engagement behaviors. Fig. 2 presents the basic framework.

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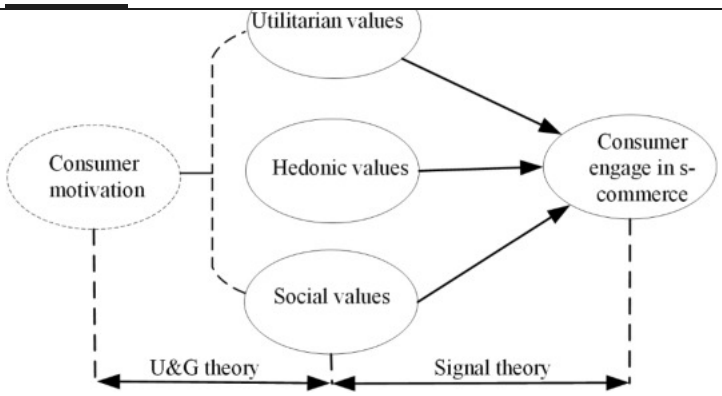

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Fig. 2. Basic theory framework.

3. Research model and hypotheses development

Using U&G theory and signal theory, we developed a research model to investigate the impacts of short-form video advertising features on customer engagement behavior (see Fig. 3). According to U&G theory, perceived utilitarian values, hedonic values, and social (symbolic) values are the main motivations for consumers to engage in short-form video advertising. Thus, performance expectancy transmits a signal of utilitarian value to consumers, entertainment transmits a signal of hedonic value to consumers, and tie strength and sales approach transmit signals of social value to consumers. We justify the hypotheses in the following sections.

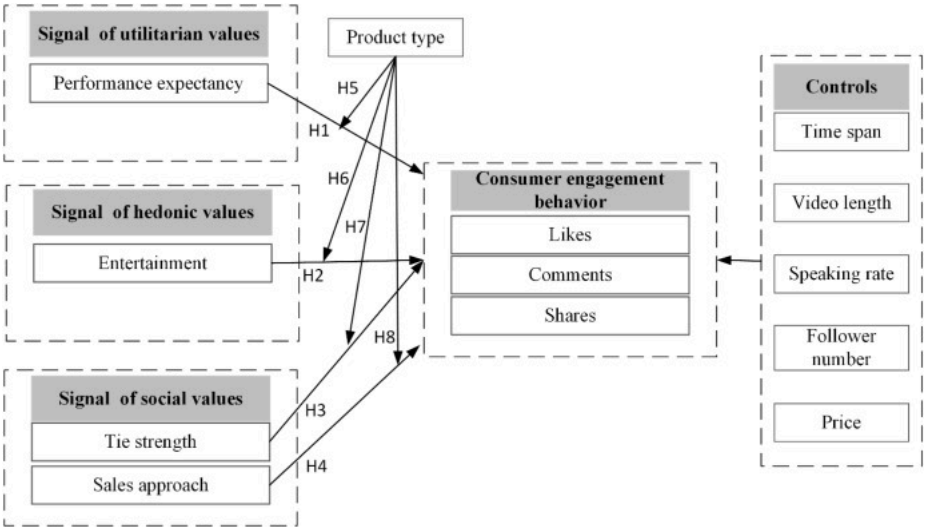

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Fig. 3. Research model.

3.1. Consumer engagement

Customer engagement can be studied in relation to two different aspects. Some researchers view customer engagement as a psychological state that occurs by virtue of consumers' interaction and co-creative experiences with agents or objects (Bolton, 2011; Li and Han, 2021), whereas other researchers focus on consumers' behaviors. In this situation, customer engagement is defined as behavioral manifestations that have a brand or firm focus, beyond purchase, resulting from motivational drivers (van Doorn et al., 2010). There is a significant difference between these two l intentions and emotions are emphasized in the psychological aspect, whereas co-creation, valence, form or nd customer goals are given greater attention from behavioral perspectives. To some extent, the latter can be ychological factors (van Doorn et al., 2010). Prior studies have mainly investigated customer engagement from lim et al., 2021; Li and Han, 2021; Molinillo et al., 2020; Wongkitrungrueng and Assarut, 2020), while ignoring the behavioral values for consumers and sellers. With the flourish of social media, consumers are able to engage with sellers more conveniently

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manifestations.

Behavior that reflects engagement with social media can be divided into three types according to the level of consumers' activeness: consumption, contribution, and creation (Muntinga et al., 2015). Consumption, such as reading or clicking, is the lowest level of engagement behavior. Contribution, such as liking, commenting, or sharing, is a more active engagement behavior, and consumers expressing their views by creating and uploading content is the most active engagement behavior (Liu et al., 2021; Schivinski et al., 2016). In short-form video platforms, such as TikTok, consumers engage mainly with sellers and others by liking, commenting, and sharing. Therefore, in our study, we focus on three active customer engagement behaviors: liking, commenting, and sharing.

3.2. Performance expectancy

Performance expectancy refers to the extent to which the customer believes using the system or product will help them meet their needs (Venkatesh et al., 2003). It could also be interpreted as other expectations, relating to usefulness, relative advantage, and outcome (Venkatesh et al., 2003). In the short-form video advertising context, sellers mainly create high-quality video content to occupy the minds of users and motivate consumers to engage with their videos (Iresearch, 2020b). When entering into short-form video advertising, consumers will be guided to pay attention to the usefulness and function of the product and consider whether the product meets their expectations. As proposed in U&G theory, if the utilitarian values of consumers can be met, the motivation of customer engagement will increase. Thus, we infer that the performance expectancy of the product will increase customer engagement behaviors. Performance expectancy has been proven to have a positive influence on customer purchase intentions in social media advertising (Alalwan, 2018). Loureiro et al. (2018) argued that performance expectancy could have a positive impact on customer satisfaction, word of mouth, and trust. Consequently, when the product in the short-form video advertisement creates a feeling of "this product is fits well with my expectancy and it is just what I need," the customer will engage more with sellers and other consumers to obtain more information. On the basis of the above discussions, we propose the following hypothesis:

H1

Performance expectancy is positively related to customer engagement behavior (H1a: likes; H1b: comments; H1c: shares).

3.3. Entertainment

McQuail (1987) argued that entertainment can satisfy audiences' needs for escapism, physical enjoyment, and emotional release. Later, Elliott and Speck (1998) referred to entertainment as the ability of an advertisement to create enjoyment and playful content and bring positive feelings to consumers. Entertainment has already been an indispensable factor in traditional markets. One of the most important functions of a short-form video platform is to provide an opportunity for users to share and enjoy interesting items with others. Unlike traditional advertisements, with short-form video advertisements, consumers always try to find interesting material to relax them. This is because short-form videos are a form of entertainment that can satisfy users' hedonic values. Therefore, we infer that entertainment can increase customer engagement behavior in short-form video advertisements. As mentioned in prior studies, entertainment has a positive impact on customer engagement with luxury social businesses (Bazi et al., 2020; Liu et al., 2021, Liu et al., 2021, Liu et al., 2021), and hedonic values induce higher levels of customer engagement in terms of knowledge sharing and social influence (Itani et al., 2020). Entertaining social media advertising with more hedonic elements could have positive influences on customer purchase intentions (Alalwan, 2018). Obviously, entertainment plays an important role in social media and social media advertising, and therefore, also in short-form video advertising. On the basis of the above discussions, we propose the following hypothesis:

H2

Entertainment has a positive impact on customer engagement behavior (H2a: likes; H2b: comments; H2c: shares).

3.4. Tie strength

Granovetter (1973) described tie strength as a combination of various factors that could build and promote relationships, such as the amount of time two or more people spend on tie interaction, their emotional intensity, and intimacy. Tie strength emphasizes a significant desire to maintain social relationships and belonging. Unlike traditional relationships, tie strength in s-commerce mainly reflects quickly formed interpersonal relationships with a seller embedded in the online business context (Ou et al., 2014). In short-form video advertisements, sellers maintain good ties with consumers by posting interesting words and replying to their comments. Frequent and timely interaction is positively related to interpersonal closeness, and interpersonal closeness tends to act as motive to protect others (Dubois et al., 2016), which means that a good tie is beneficial for trust and credibility (Hershkovitz and Hayat, 2020). When consumers' trust in sellers increases, they may be more willing to engage with sellers. In addition, sellers' replies can increase the sense of belonging of consumers. This is because individuals tend to feel supported and recognized when interacting with others. After obtaining support and replies from the seller, the customer will be more willing to maintain this gage more with sellers (Hampshire et al., 2017; Kang et al., 2021). On the basis of the above discussion, we

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customer engagement behavior (H3a: likes; H3b: comments; H3c: shares).

The sales approach describes the interaction and value production between sellers and consumers. The marketing and selling literature discusses two extreme sales approaches: transaction based and relationship based (Anderson and Narus, 1991). In the transaction-based approach, sellers focus almost all the time on selling their products and discussing the product and price. Thus, they can build only simple relationships with consumers in the short term (Wongkitrungrueng and Assarut, 2020). However, in the relationship-based approach, sellers share their private lives, create customer values, and treat consumers as friends and relatives (Inks et al., 2019; Wongkitrungrueng and Assarut, 2020). In general, the difference between these two approaches is that sellers who use a relationship-based approach focus more on sharing their lives and building closer relationships, as well as creating value for consumers and their followers, whereas those who use a transaction-based approach do not. However, these two approaches are not mutually exclusive; rather, they are a phase in an evolutionary process. Thus, they can be pursued simultaneously with different combinations (Lefaix-Durand and Kozak, 2009).

Unlike traditional e-commerce models, consumers have different attitudes and opinions regarding short-form video advertisements. Specifically, besides the attributes of traditional models, consumers pay attention to the social attributions of the short-form video advertisement, which can be regarded as unique features of short-form video advertising. Users engage in short-form videos for gratification and out of narcissism (Fei et al., 2021; Hu et al., 2016). In short-form video advertisements, if sellers promote their products in the same way as on traditional e-commerce platforms, this will result in consumer fatigue. To some extent, the development of Web 2.0 has caused serious information overload. Excessive marketing information makes consumers resistant, which renders marketing ineffective (Iresearch, 2020a).

According to U&G theory, only when their values can be met will the seller become engaged. In this situation, the transaction-based sales approach may decrease consumer engagement behavior in relation to short-form video advertisements, whereas the relationship-based sales approach may encourage consumer engagement behavior. On the basis of the above discussion, we propose the following hypothesis:

H4

The higher the degree to which the transaction-based sales approach is used, the less the consumer will engage in short-form video advertisement (H4a: likes; H4b: comments; H4c: shares).

3.6. Moderating effect of product type

Nelson (1970) proposed that goods can be divided into two types: experience goods and search goods. Experience goods are products for which it is difficult for consumers to search for information concerning their quality, while search goods are products that are convenient to search and that can be evaluated in terms of quality by the information available (Lee and Shin, 2014; Nelson, 1970). In fact, all goods involve a combination of search and experience attributes (Alba et al., 1997; Lynch Jr and Ariely, 2000). Search attributes (e.g., price) are objective, and easy to compare, whereas experience attributes (e.g., how easy a product is to use) are equivocal, and difficult to evaluate (Huang et al., 2009). These differences can change the way consumers process information. Therefore, we propose that product type can have a moderating effect on the influence of short-form video advertising on customer engagement.

Studies in the literature have focused on the influence of experience attributes on customer cognition and behaviors (Benoit et al., 2019; Huang et al., 2009; Lee and Shin, 2014; Lian and Yen, 2013; Park and Lee, 2009; Senecal and Nantel, 2004). In short-form video advertisements, sellers earn a commission from the producers and initial sellers, mainly by sharing their product experience. Performance expectancy directly reflects the function and usefulness of the product. Unlike for experience goods, this feature can be judged more easily in searches; therefore, it is more convenient for consumers in making decisions and engaging with sellers. Therefore, we infer that performance expectancy has stronger impacts on engagement behavior for search goods than for experience goods.

Consumers engage in more heuristic thinking for experience goods (Axsom et al., 1987; Lee and Shin, 2014). Entertainment mainly reflects efforts that the seller makes regarding the video and the seller's attractiveness; thus, they may consider its use to be more acceptable in experience goods than in search goods. In this situation, we propose that entertainment is more strongly related to customer engagement behavior for experience goods than for search goods. Similarly, tie strength mainly reflects the intimacy of the relationship between sellers and consumers. Consumers may judge sellers' trust and credibility through tie strength (Hershkovitz and Hayat, 2020) and thereby be more willing to engage with sellers for experience goods than for search goods. Therefore, we propose that tie strength has greater effects on engagement behavior for experience goods than for search goods. When sellers take more effort introducing the price and promoting the product, it will reduce consumers' desire to engage with sellers for experience goods rather than for search goods. This is because, unlike for search goods, there is greater uncertainty for consumers in evaluating experience goods, and a transaction-based sales approach may result in a greater sense of low quality and consumer fatigue for experience goods than for search goods. Thus, we propose that the sales approach has a greater influence on engagement behavior for experience goods than for search goods.

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tancy on consumer engagement behavior is greater for search goods than for experience goods (H5a: likes;

H6

comments, H8c: shares).

H7

The influence of tie strength on consumer engagement behavior is greater for experience goods than for search goods (H7a: likes; H7b: comments; H7c: shares).

H8

The influence of sales approach on consumer engagement behavior is greater for experience goods than for search goods (H8a: likes; H8b: comments; H8c: shares).

3.7. Control variables

Since our focus is the feature of short-form video advertising that can influence consumer engagement behavior, we control for the impacts of other factors that may also influence consumer engagement behavior. Time span refers to the history of the video since its release. The videos we collected were posted at different times and videos posted earlier may have more likes, comments, and shares; thus, we controlled its influence. Video length is the duration of a video. According to a report (feigua, 2021), the duration of short-form video advertisements can affect video performances and consumer engagement behavior; therefore, we control this variable. Speech rate is one of the most important phonetic features, reflecting the amount of information and knowledge about a product (Liu et al., 2020); thus, we control its influence on consumer engagement behavior. The follower number is the number of users who have followed the seller, which is an important indicator of sellers' reputations (Kromidha and Li, 2019; Liu and Park, 2015). Since sellers' reputations may influence customer behavior and decision-making (Dai et al., 2018; Van Der Heide et al., 2013), we controlled this variable. The price of the product reflects the product value, which may influence consumers' satisfaction and choice behaviors (Cai and Xu, 2006); thus, this was also controlled in this study.

4. Research methods

We provided a two-step framework for the data analysis, as shown in Fig. 4. In the first step, we transformed the videos into text and cleaned all the textual data. In the second phase, we calculated the values of each variable for all samples.

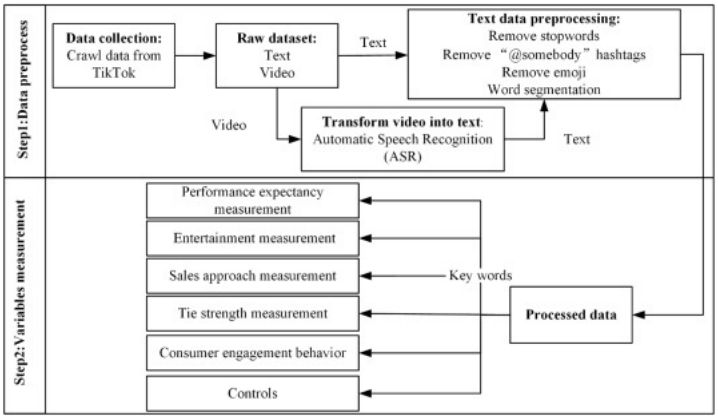

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Fig. 4. Data analysis process.

4.1. Research site

We chose TikTok as the source of data to examine the impacts of short-form video advertising features on consumer engagement behavior for three reasons. First, according to research (199IT, 2021), total global downloads on TikTok have exceeded three billion. The number of daily active users of TikTok has exceeded 600 million and the average number of daily video searches has exceeded 400 million in China (TikTok, 2021), making TikTok one of the world's largest mobile short-form video-sharing platforms. This large volume of users enabled us to obtain more data. Second, unlike other platforms, such as Little Red Book and BiliBili, which provide different forms of videos and posts, TikTok only supports short-form video and livestreaming. Consequently, sharing and enjoying short-form video is the most popular activity for TikTok users, which means more in short-form videos than in posts or livestreaming. Finally, TikTok has more e-commerce partners, cts in short-form video advertisements. In general, TikTok provided us with more comprehensive and accurate ate platform for our study.

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4.2. Data collection and transformation

Article Recommended Articles Cited by (124) Metrics

Depending on whether consumers can evaluate a product before purchase, Nelson (1970) classified products into search products and experience products. This classification has been widely adopted in consumer research (Girard and Dion, 2010). Search products, such as digital products, furniture, and sporting equipment, can be easily appraised by comparing products' attributes (Jiménez and Mendoza, 2013). Experience products, such as cosmetics, food, and liquor, are more difficult for consumers to obtain information on, or to compare product attributes for, because consumers usually need to physically see, smell, touch, or use them (Jiménez and Mendoza, 2013), and different consumers are likely to form heterogeneous opinions on the basis of a single product attribute (Hong and Pavlou, 2014). We chose cosmetics to represent experience products and digital goods to represent search products in this study. We entered two keywords (i.e., "cosmetics" and "digital goods") in the TikTok APP search box. Then, we obtained the search results for cosmetics and digital goods, respectively.

In consideration of the impact of the active period of videos on data, keeping the validity of variables, and reducing error, we collected videos that were posted 30 days or more prior to the data collection date. Finally, data on a total of 1000 videos and about three million reviews posted by the 27 top sellers over the period from January 2020 to June 2021 were collected using Python programming language combined with Appium automation technology and the Fiddler4 web debugging tool.

The raw data we collected were unstructured, so we needed to transform them into the structured form. First, we deleted invalid data with missing information. Second, text analysis technology was applied for data cleaning; we cleaned "@somebody" hashtags from each comment because consumer identification would pose serious problems for text analysis. We extracted video URLs from the raw data and downloaded them in the form of MP4. After that, we extracted audio files in "wav" format from the videos, and then, Baidu voice-recognition API was employed to convert the speech into text. Baidu voice-recognition API was developed by the Baidu artificial intelligence open platform, and its accuracy rate is higher than 90%. Next, we removed stop words from all comments and video texts. Then, we used Jieba to perform word segmentation. Jieba is a third-party module for text processing, which can accurately segment Chinese text into words. Finally, a total of 800 samples was saved.

4.3. Measures

4.3.1. Performance expectancy and entertainment

Variables of performance expectancy and entertainment were identified from the comments of videos posted by viewers because there were no readily available metrics to be used as proxies. Therefore, the qualitative textual data were transformed into quantitative data to measure performance expectancy and entertainment.

To code whether or not a comment contained textual components representing performance expectancy and entertainment, identifying a group of words that adequately portrayed the two variables was critical. We first built a Latent Dirichlet Allocation model to explore the topic distribution of all comments. The results showed two obvious topics related to performance expectancy and entertainment. We extracted some words (i.e., performance expectancy—"useful" and "product effect"—and entertainment—"beautiful" and "interesting") as initial keyword dictionaries for performance expectancy and entertainment. We then selected other words that denoted performance expectancy and entertainment from the word segmentation of all comments. Finally, 48 words (including "cost performance," "ingredient," and "coupon") and 50 words (such as "strong," "admire," and "fairy"), encompassing nouns, verbs, adjectives, and adverbs, were selected to identify comments that represented performance expectancy and entertainment, respectively.

We counted the total number of comments containing the keywords in these two groups respectively as measurements for performance expectancy and entertainment.

4.3.2. Sales approach

To measure the sales approach, we analyzed the texts of sellers' speech, which were converted by Baidu voice-recognition API. We started with the two words "to buy" and "hurry up to order", which are salient in traditional transaction-based sales approaches (Wongkitrungrueng and Assarut, 2020). We then selected the words that directly illustrate product promotion (i.e., "discount," "highly recommend," and "must-have products") from the speech texts. Finally, a group of words was obtained that could represent the transaction-based sales approach. We used the ratio of the number of words contained in the group to the total number of words in sellers' speeches to measure the sales approach.

4.3.3. Tie strength

Tie strength in s-commerce mainly reflects the quickly formed interpersonal relationship between consumers and sellers (Ou et al., 2014). Friendly [illegible] nsumers can quickly build good ties between them. In this study, we used the total number of replies between [illegible] rement of tie strength; this was also used in Kang et al. (2021).

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[illegible] t behavior

of likes, comments, and shares to measure consumer engagement behavior.

4.3.5. Control variables

We used the number of days from the date the video was posted to the time the video was collected to measure the time span, and video duration to measure the video length (the video length is generally between 30 and 300s). Speech rate was measured by the number of words contained in the unit time (e.g., if a 40s video contained 160 words, the speech rate was four words per second). The follower number was measured by the number of fans. Product price was measured by the price shown in the video. The measurements of all variables are shown in Table 2. Descriptive statistics of the measurements are shown in Table 3.

Table 2. The measurements of all variables.

Variables		Definition	Measurements
Independent variables	Performance expectancy	The degree of usefulness and outcome expectations of products or service	The total number of comments that discuss the product
	Entertainment	The ability to satisfy audiences' needs for escapism, physical enjoyment, and emotional release	The total number of comments that contain playful words and the description of physical attraction about the seller
	Tie strength	A quickly formed interpersonal relationship with the seller embedded in the online business context	The total number of replies between the seller and consumers
	Sales approach	The interaction and value-producing between the seller and customer	The ratio of the number of words that contain the meanings of direct promotion to the total number of words
Control	Time span		The days from the date the video was posted to the date the video was collected
	Video length		Video duration
	Speech rate		The number of words contained in unit time
	Follower number		The number of fans
	Price		The price shown in the video
Dependent variables	Consumer engagement behavior	Consumer's behavioral manifestations that have a brand or firm focus, beyond purchasing, resulting from motivational drivers	The total numbers of likes, comments, and shares of the video

Table 3. Descriptive statistics of data for the paper.

Variables	Observation	Min	Max	Mean	Std. Dev.
Performance-expectancy	800	1.00	5326.92	301.66	583.26
Entertainment	800	0.00	1900.76	56.79	147.52
Tie strength	800	3.00	14,349.00	437.48	993.87
Sale approach	800	0.00	0.06	0.01	0.01
Time span	800	1.00	533.00	128.20	107.65
Video length	800	8.61	893.90	104.00	89.23
Speech speed	800	0.00	25.88	5.8	2.28
Follower number	800	426,018.00	13,639,079.00	3,827,167.16	3,031,303.68
Price	800	10.00	23,999.00	1334.01	2640.06
	800	281.00	1,564,100.00	84,445.97	133,210.05
	800	20.00	44,516.00	2461.89	4890.30
	800	4.00	40,900.00	1057.77	2820.45

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4.4.1. Data validity check

Several assumptions should be met when we use multiple linear regression: continuous response variance, linearity, normal distribution of residuals, homoscedasticity, and no multicollinearity (Tranmer and Elliot, 2008).

First, the measurements of dependent variables are the number of likes, comments, and shares. They are all continuous variables. Second, to solve the problem of high skewness and kurtosis and non-normal distribution, the logarithmic non-linear transformation by adding 1 to avoid the possibility of taking a log of 0 was performed (Shahbaznezhad et al., 2021). Then, a multicollinearity test was performed, showing that there was no multicollinearity between the independent variables. The results indicated no serious multicollinearity between independent variables since the variance inflation factor was below 5. Finally, the measure of logarithmic transformation could reduce the endogeneity of independent variables. We also controlled the impact of time span, video length, speaking rate, follower number, and price to reduce endogeneity.

4.4.2. Regression analysis

The multiple linear regression models were developed as follows:

$$lnlikes = \beta_0 + \beta_1 lnPE + \beta_2 lnENT + \beta_3 lnTS + \beta_4 lnSA + \beta_5 lnPE * TYPE + \beta_6 lnENT * TYPE + \beta_7 lnTS * TYPE + \beta_8 lnSA * TYPE + \beta_9 lnTISP + \beta_{10} lnVL + \beta_{11} lnSR + \beta \quad (1)$$

$$lncomments = \beta_0 + \beta_1 lnPE + \beta_2 lnENT + \beta_3 lnTS + \beta_4 lnSA + \beta_5 lnPE * TYPE + \beta_6 lnENT * TYPE + \beta_7 lnTS * TYPE + \beta_8 lnSA * TYPE + \beta_9 lnTISP + \beta_{10} lnVL + \beta_{11} ln \quad (2)$$

$$lnshares = \beta_0 + \beta_1 lnPE + \beta_2 lnENT + \beta_3 lnTS + \beta_4 lnSA + \beta_5 lnPE * TYPE + \beta_6 lnENT * TYPE + \beta_7 lnTS * TYPE + \beta_8 lnSA * TYPE + \beta_9 lnTISP + \beta_{10} lnVL + \beta_{11} lnSR + \quad (3)$$

In Equations (1), (2), (3), *lnlikes*, *lncomments*, and *lnshares* were dependent variables; *lnTS*, *lnPE*, *lnENT*, and *lnSA* were independent variables; *lnTS * TYPE*, *lnPE * TYPE*, *lnENT * TYPE*, and *lnSA * TYPE* were interactions; *lnTiSp*, *lnVL*, *lnSR*, *lnFN*, and *lnP* were controls; TYPE was the dummy variable; and ε_i, ε_j, and ε_k were random errors.

5. Results

The results for the multivariate linear regression are shown in Table 4. This provides evidence that performance expectancy was positively related to consumer engagement behavior in the comments (β=0.337, p<0.01), which supports H1b. However, there was no significant influence of performance expectancy on likes and shares, which means that H1a and H1c are rejected. We found that entertainment had significant positive influences on likes (β=0.165, p<0.01), comments (β=0.309, p<0.01), and shares (β=0.323, p<0.01), supporting H2a, H2b, and H2c, respectively. Tie strength was found to have positive influences on likes (β=0.152, p<0.01), comments (β=0.323, p<0.01), and shares (β=0.237, p<0.01), which supports H3a, H3b, and H3c. Sales approach was found to have significant negative influences on likes (β=−0.066, p<0.05), comments (β=−0.058, p<0.05), and shares (β=−0.095, p<0.01), supporting H4a, H4b, and H4c.

Table 4. Multivariate linear regression result.

Variables		ln (likes)	ln (comments)	ln (shares)
Independent variables	ln (performance expectancy)	−0.075	0.337**	0.097
	ln (entertainment)	0.165**	0.309**	0.323**
	ln (tie strength)	0.152**	0.323**	0.237**
	ln (sales approach)	−0.066*	−0.058*	−0.095**
Interaction	ln (performance expectancy)_Type	0.472**	−0.111	0.158
	ln (entertainment)_Type	0.008	−0.086*	−0.091
	ln (tie strength)_Type	0.215	0.250*	0.194
	ln (sales approach)_Type	0.013	0.055*	0.022
	ln (time span)	−0.002	0.009	0.174**
	ln (video length)	−0.158**	0.032	0.106**
	ln (speech rate)	0.002	0.003	0.027

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ln (follower number)	0.408**	0.078**	0.160**
ln (price)	−0.062*	0.016	0.085**
Type	−1.050**	−0.08	−0.323**
Constant	1.664*	0.288	−3.489**
Observations	800	800	800
Adjust R-squared	0.719	0.784	0.566

Note:**p<0.01, *p<0.05.

As for control variables, the time span was found to have a significant positive effect on share behaviors (β=0.174, p<0.01), but had no effects on likes and comments. Moreover, video length was found to have an obvious negative impact on likes (β=−0.158, p<0.01), and a positive influence on shares (β=0.106, p<0.01). This indicated that a longer video would increase the possibility of sharing behavior among consumers, but could have a negative impact on likes. Speech rate was found to have no significant effects on consumer engagement behavior. The number of followers provided significant positive effects on likes (β=0.408, p<0.01), comments (β=0.078, p<0.01), and shares (β=0.16, p<0.01). Finally, the price of the product was found to have a negative impact on likes (β=−0.062, p<0.05), but a positive impact on shares (β=0.085, p<0.01).

We also applied group regression to check the moderating effects of product type. The results are shown in Table 5. Notably, there is no significant difference between the results of Table 4 and those of Table 5.

Table 5. Moderating effect of product types.

Variables	ln (likes)		ln (comments)		ln (shares)	
	b0-b1	Z-value	b0-b1	Z-value	b0-b1	Z-value
ln (performance_expectancy)	−0.465	−4.30**	0.103	1.486	−0.207	−2.22**
ln (entertainment)	0.076	0.851	0.129	2.280*	0.109	1.581
ln (tie strength)	−0.036	−0.417	−0.202	−3.482**	−0.011	−0.16
ln (sales approach)	−0.032	−0.566	−0.078	−2.02*	−0.041	−0.804

Note: **p<0.01, *p<0.05.

First, the results in Table 5 revealed that the effect of performance expectancy on likes was greater for search goods than for experience goods (β =0.472, p<0.01), supporting H5a. The effect of performance expectancy on comments and shares showed no difference between experience goods and search goods, rejecting H5b and H5c. Second, we discovered that the effect of entertainment on comments was stronger for experience goods than for search goods ($\beta = -0.086$, p<0.01), which supported H6b. The effect of entertainment on likes and shares showed no difference between experience goods and search goods; thus, H6a and H6c were rejected. Third, the result revealed that tie strength had a greater effect on comments for search goods than for experience goods ($\beta = 0.25$, p<0.01), supporting H7b. The effect of tie strength on likes and shares showed no significant difference between experience goods and search goods; thus, H7a and H7c were rejected. Finally, we found that the effect of sales approach on comments was stronger for experience goods than for search goods (β=0.055, p<0.01), supporting H8b, and the effect of sales approach on likes and shares showed no apparent difference between experience goods and search goods, rejecting H8a and H8c.

Fig. 5 shows the interaction pattern. Fig. 5a shows the relationship between performance expectancy and likes for experience goods and search goods respectively. As expected, the line of search goods was steeper than the line of experience goods, which means that performance expectancy had a stronger effect on likes for search goods than for experience goods. Fig. 5b shows the relationship between entertainment and comment behavior for experience goods and search goods. The figure demonstrates that the line for experience goods was steeper than the line for search goods, which means that entertainment had a stronger effect on likes for search goods than for experience goods. Fig. 5c shows the relationship between tie strength and comment behavior. The line for search goods was more steeply sloped than that for experience goods, indicating that the was greater for search goods than for experience goods, and confirming H7b. Finally, in Fig. 5d, the effect of the tronger for experience goods than for search goods since the line for search goods was more steeply sloped than

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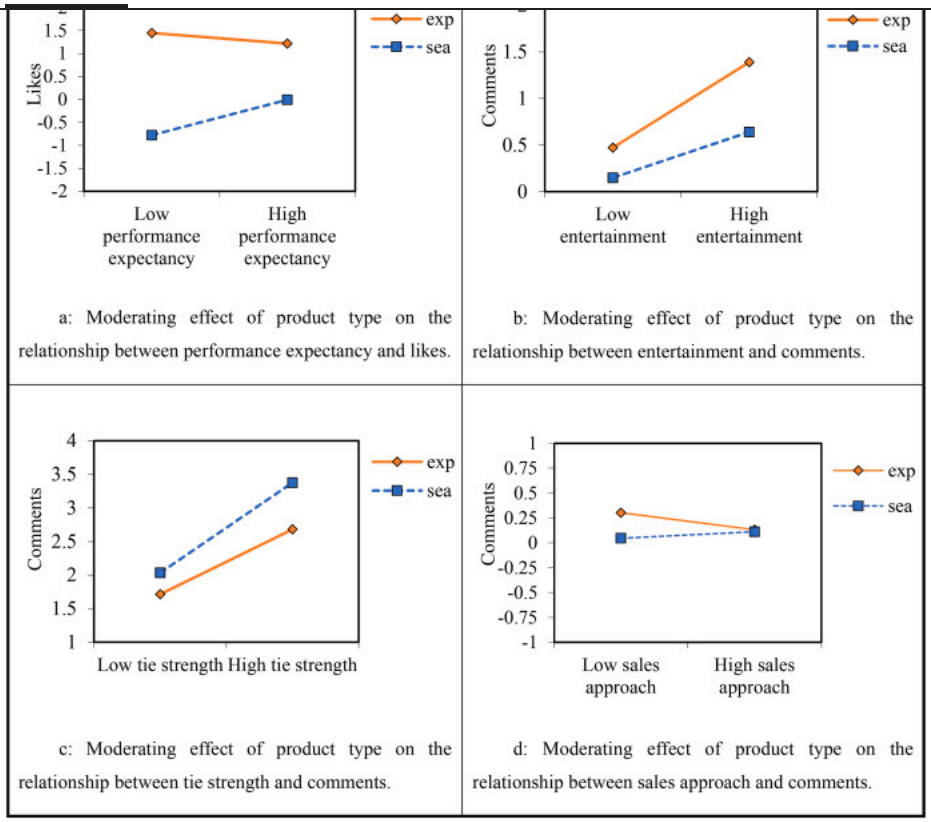

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Fig. 5. Moderating test result.

6. Discussion and implications

6.1. Discussion of key findings

There are several findings from the results. First, performance expectancy was found to have a significant positive influence on consumers' comment behavior. Most studies related to performance expectancy have focused mainly on the impact of performance expectancy on purchase intention, consumer satisfaction, and electronic word of mouth (Alalwan, 2018; Loureiro et al., 2018), ignoring the influence of performance expectancy on consumer engagement behavior. Our findings extend the results of earlier studies by investigating the effect of performance expectancy on real consumer engagement behavior.

Second, entertainment was found to have positive impacts on likes, comments, and shares. This is consistent with results from Liu et al. (2021), who stated that entertainment is an essential element in social media and s-commerce. Clearly, users engage in social media partly to escape from unpleasant and pressured realities. Entertainment can also reduce the degree of information overload in videos, and thereby allow consumers to receive the information more efficiently.

Third, tie strength was found to have positive effects on likes, comments, and shares. While Kang et al. (2021) focused on the positive role of tie strength on consumer engagement behavior in livestreaming commerce and Nanne et al. (2021) explored the moderating effect of tie strength between senders' facial expressions and consumers' engagement intentions as indicated by likes, comments, and purchases in visual social network sites, our study proved the positive influence of tie strength on consumer engagement behavior in the field of short-form video advertising.

Fourth, the sales approach proved to have a negative impact on consumer engagement behavior. This finding is in line with Menon et al. (2019), nt has a negative effect on consumer engagement. However, Wongkitrungrueng and Assarut (2020) revealed pach is beneficial for building a short-term relationship between sellers and consumers in livestreaming their results. This may be because, unlike short-form video commerce, consumers in livestreaming commerce d therefore, become more concerned about the product.

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performance expectancy on likes is stronger for search goods than for experience goods and the influence of entertainment and sales approach on comments is stronger for experience goods than for search goods. This finding is in line with Axsom et al. (1987) and Lee and Shin (2014), who stated that consumers use more heuristic thinking regarding experience goods than search goods. Apparently, consumers are more willing to accept performance expectancy in regard to search goods because it is easier to make judgments, whereas they consider that entertainment and sales approaches are more important for experience goods. However, the results indicated that tie strength has a stronger influence on comments about search goods than experience goods. This may be because it is easier for consumers to understand and communicate with sellers about various topics surrounding search goods, whereas for experience goods, consumers focus more on asking questions and thereby interact less with sellers. By exploring the moderating effect of product type on the relationship between these four factors and consumer engagement behavior, the study emphasized the important role of product type for research on consumer engagement behavior.

6.2. Theoretical implications

There are several theoretical implications from this research. First, this study contributes to the social media advertising literature by examining a new type of advertising—short-form video advertising. Studies about social media advertising have mainly focused on post advertising or livestreaming advertising (Kang et al., 2021; Liu et al., 2021; Wongkitrungrueng and Assarut, 2020), while ignoring the newer short-form video advertising. Conversely, although the social role of short-form video has received the most attention (Fei et al., 2021; Hu et al., 2016; Zhang et al., 2019), its commercial roles have not been explored effectively. This study is one of the first to investigate the impact of short-form video advertising on consumer engagement behavior, thereby enriching the social media advertising literature.

Second, this study extends the research on consumer engagement behavior by applying a combination of U&G and signal theories. Prior studies have mainly applied U&G theory to investigate the motivation of consumer engagement from the perspective of consumers, while ignoring real consumer engagement behavior manifestations (Bazi et al., 2020; Busalim et al., 2021; Molinillo et al., 2020; Verhagen et al., 2015). Studies exploring factors that influence the manifestation of consumer engagement have paid little attention to consumers' opinions and motivations (Liu et al., 2021; Shahbaznezhad et al., 2021). Our study considered both the motivations for consumer engagement and its manifestations, which could make the search results of consumer engagement behavior closer to the real situation. From this perspective, our study makes contributions to the research on consumer engagement behavior.

Third, this study highlights the significance of product type in the advertising literature. Existing studies have mainly explored the moderating effect of product type on the relationship between online consumer reviews and purchase intentions (Lee and Shin, 2014; Lian and Yen, 2013; Park and Lee, 2009). Although our research content differs significantly from those studies, consumers process information in a similar way. Consumers are likely to concentrate more on product experience for experience goods than for search goods. The findings of this study indicate that for different types of products, different advertising strategies are needed to attract consumers. Future research exploring consumers' behavior should also consider the difference between various product types.

Finally, this study contributes to social media research from a methodological perspective. Existing studies have mainly used the survey method to explore users' motivations, attitudes, and behaviors (Fei et al., 2021; Rubenking, 2019; Yiyi et al., 2022). The biggest restrictions of the survey method are small sample sizes, the recall bias of respondents, and the limited questions presented in surveys. This study illustrates that the use of big data such as data spiders and text mining can effectively overcome the restrictions of the survey method, which in turn can provide more objective results. Future research examining user behavior in other social media-related contexts could also adopt this methodology to obtain more reliable and objective results.

6.3. Practical implications

This research also has some practical implications for sellers and consumers, and managerial implications for strategic marketers. For sellers, first, because the result revealed that performance expectancy has a positive effect on comments, sellers should pay attention to introducing their product's performance according to consumers' expectancy in their videos. For instance, sellers can analyze consumers' product needs and preferences through data mining.

Second, entertainment was found to have positive effects on likes, comments, and shares. Thus, sellers should increase the element of entertainment in their short-form video advertisements. Short-form video platforms such as TikTok have promoted many entertainment elements for users to create their own playful content, so sellers can use playful pictures, emojis, and texts in their videos, as well as physical attraction, to increase entertainment.

e positive effects on consumer engagement behavior. Therefore, the seller should facilitate tie strength with
answer consumers' questions punctually and interact positively with consumers in comments by including
seful. A sales approach was found to have negative impacts on consumer engagement behavior. This means that
sed sale approaches and avoid using words that strongly recommend consumers purchase their product or
untrustworthy words, such as "to buy," "to order," "very effective," "discount," "hurry up," "used up," and other related words.

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consumer engagement behavior, sellers should consider the role of product type when promoting different types of products. For instance, sellers who promote search goods should focus more on product performance for more likes, in contrast to those who promote experience goods. Further, they should focus on tie strength for more comments, whereas entertainment can be limited, because excessive entertainment does not cause an apparent increase in consumer engagement for search goods. For experience goods, more entertainment is needed to increase comments than for search goods, and sellers also need to avoid using a sales transactional approach. In particular, sellers should pay attention to the video length since longer videos increase shares. They should also engage in effective measures to increase their numbers of followers and avoid promoting expensive products.

For consumers, the degree of consumer engagement behavior shows the ability of the video to meet their motivations and personal needs. Poor consumer engagement behavior represents a video's low quality. The number of followers can reveal information for consumers. More fans represent higher reputation and trust.

Our study helps strategic marketers better evaluate the marketing effectiveness of short-form video advertising and enables them to make faster forecasting. Specifically, these four factors in our study have been proved to have significant impact on consumer engagement behavior, and managers could estimate the marketing effect of short-video advertisements on the basis of these.

Second, our study helps marketing managers make better decisions when searching for advertisers to promote their products. With a growing number of short-form video users emerging to promote products, managers are facing challenges choosing the advertisers suitable for products promotion. These findings of the four factors (i.e., performance expectancy, entertainment, tie strength, and sales approach) and product type in our study provide marketing managers with basic standards, which could bring marketing managers reliable data-driven decisions.

7. Limitations and conclusion

7.1. Limitations

The current research has some limitations. First, we only investigated the user groups of TikTok. User groups of other social platforms (e.g., Little Red Book and BiliBili) were not included. Unlike TikTok, other short-form video platforms have different demographic characteristics (for instance, users of Little Red Book are generally young people), which may lead to different results. It would be interesting to replicate this study on these two platforms to compare the similarities and differences in the results.

Second, we only examined a limited number of factors that may influence consumer engagement in this study. Some frame features, such as height–width ratio, feature complexity, facial features, human body features, spatial distance (Tok et al., 2021), camera view (Wang, 2020), and voice characteristics (Liu et al., 2020), may also influence consumer engagement behavior. Future research could explore the influences from these factors. Moreover, through a macroscopic lens, as employed in the studies of Aazami and Saidi-Mehrabad (2021) and Alaei and Manavizadeh (2022), it would be interesting to explore the game process among consumers, sellers, and manufacturers.

7.2. Conclusion

Triggered by the booming of short-form video advertising, this study attempts to gain a deeper understanding of consumer behavior when engaging with short-form video advertisements. U&G and signal theories were used as the theoretical framework through which to examine the effects of short-form video features on consumer engagement. By analyzing data obtained from TikTok, this study found that performance expectancy has a significant positive influence on consumers' comments; likewise, entertainment, tie strength, and the sales approach have significant positive influences on likes, comments, and shares. Moreover, product type moderates the relationship between these four features and consumer engagement behavior. Specifically, the effect of performance expectancy on likes is stronger for search goods than for experience goods; conversely, the influences of entertainment and sales approach on comments are stronger for experience goods than for search goods, while tie strength has a greater influence on comments for search goods than for experience goods. This study contributes to the social media advertising literature by examining a new type of advertising. It extends the findings of the short-form video literature by uncovering how short-form video features can influence users' engagement behaviors. Moreover, it highlights the significance of product type in the advertising literature by demonstrating how product type can moderate the impact of short-form video features on user engagement. From a methodological perspective, this study illustrates how user data on short-form video platforms can benefit researchers and help them to explore users' behaviors on these platforms. From a practical perspective, this study provides valuable suggestions for sellers who promote products through short-form video platforms and strategic digital marketers.

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Reading Assistant